Chapter 5

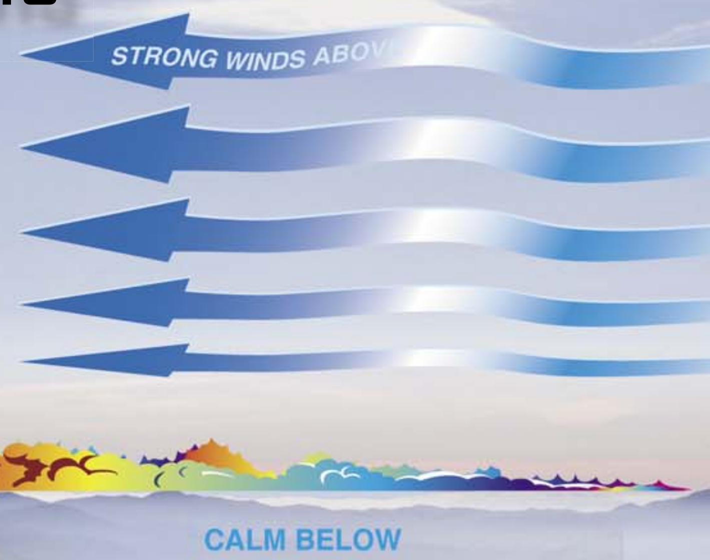
Preflight and Ground Operations

Introduction

Preflight preparations should include the overall evaluation of the:

- **P**ilot: experience, sleep, food and water, drugs or medications, stress, illness
- **A**ircraft: certificates/documents, airworthiness, fuel, weight (does not exceed maximum), performance requirements, equipment
- En**V**ironment: weather conditions, density altitude, forecast for departure and destination airfields, route of flight, runway lengths
- **E**xternal pressures: schedules, available alternatives, purpose of flight

Often remembered as PAVE, it is important to consider each of these factors and establish personal minimums for flying.



Where To Fly

The weight-shift control (WSC) aircraft can be transported by trailer from one flying field to the next. For as many benefits as this provides, transporting the aircraft into unfamiliar territory also includes some safety and operational issues.

Contact airport management to inquire about any special arrangements to be made prior to arriving by trailer *[Figure 5-1]* and there may be special considerations for flying WSC aircraft with other aircraft. With smaller patterns typically used by WSC aircraft, as covered in Chapter 10, Airport Traffic Patterns, airport management may want a pilot to operate over sparsely populated areas rather than the normal airplane patterns over congested areas because of the unique noise of the WSC aircraft. *[Figure 5-2]* Check the Airport/Facility Directory (A/FD) all required airport information per Title 14 of the Code of federal Regulations (14 CFR) part 91 section 103, Preflight information. Some operation examples are traffic pattern information, noise abatement procedures , no fly zones surrounding the airport, and special accommodations that may need to be arranged for WSC aircraft..

Because of the wide range of flying characteristics of the WSC aircraft, inform local pilots about some of the incidentals of the specific WSC aircraft (e.g., flying low and slow for certain configurations). The more non-WSC aircraft pilots know about WSC flight characteristics and intentions, the better they understand how to cooperate in flight. Sharing the same airspace with various aircraft categories requires pilots to know and understand the rules and understand the flight characteristics and performance limitations of the different aircraft.

Figure 5-2. *Contact local airport management to determine best operation for the aircraft and its type of operation.*

Figure 5-1. *Contact the local airport management to find an acceptable location to stay at the airport.*

For operations at nonaircraft fields, special considerations must be evaluated. Permission is necessary to use private property as an airstrip. Locate the area on an aeronautical sectional chart to check for possible airspace violations or unusual hazards that could arise by not knowing the terrain or location. Avoid loitering around residential structures and animal enclosures because of the slow flight characteristics of WSC aircraft and distinct engine noise.

While selecting a takeoff position, make certain the approach and takeoff paths are clear of other aircraft. Fences, power lines, trees, buildings, and other obstacles should not be in the immediate flightpath unless the pilot is certain he or she is able to safely clear them during takeoff and landing operations.

Walk the entire length of the intended takeoff and landing area prior to departure. *[Figure 5-3]* Look for holes, muddy spots, rocks, dips in the terrain, high grass, and other objects that can cause problems during takeoff and landing. Physically mark areas of concern with paint, flags, or cones. Uneven ground, mud, potholes, or items in fields such as rocks might not be visible from the air. Plowed rows and vegetation are larger than they appear from the air. Unfamiliar fields can make suitable landing areas for emergencies, but should not be used as intended landing areas. Extreme caution must be exercised when operating from a new field or area for the first time.

Figure 5-3. *Fields that look like good landing areas from the air may actually be hazardous.*

Preflight Actions

A pilot must become familiar with all available information concerning the flight, including runway lengths at airport of intended use, takeoff and landing distance accounting for airport elevation and runway slope, aircraft gross weight, wind, and temperature. For a cross-country flight not in the vicinity of the takeoff/departure airport, information must include weather reports and forecasts, fuel requirements, and alternatives available if the planned flight cannot be completed.

Weather

Weather is a determining factor for all flight operations. Before any flight is considered, pilots should obtain regional and local information to first determine if the predicted weather for the planned flight is safe.

Regional Weather

Understanding the overall weather in the region being flown provides an overview of conditions and how they can change during flight. Fronts, pressure systems, isobars, and the jet stream determine the weather. There are a number of information resources from which to find the regional view of weather systems, observed and predicted. Surface analysis charts show these regional systems, which are common on weather internet sites and TV broadcasts. *[Figure 5-4]* Review the Pilot's Handbook of Aeronautical Knowledge for a comprehensive understanding of weather theory, reports, forecasts, and charts for weather concepts covered throughout this weather section.

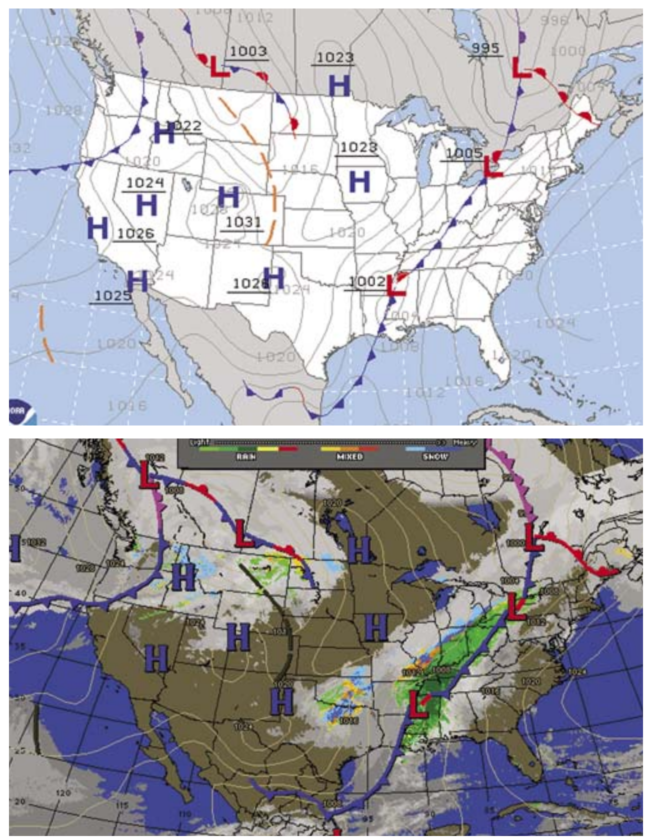


Figure 5-4. *Standard surface analysis showing fronts, pressure systems, and isobars (top) and composite surface analysis which adds radar and infrared satellite to show cloud cover (bottom).*

There are many sources for obtaining a weather briefing, such as www.aviationweather.gov, www.nws.noaa.gov, 1-800-WX-BRIEF, and a variety of internet sites that specialize in local and regional weather.

Local Conditions

In gathering weather information for a flight, obtain current and forecast conditions where flying, as well as alternate airports in case landing at the intended destination is not possible. These conditions should include wind (surface and winds aloft), moisture, stability, and pressure.

Surface wind predictions and observations can be looked at with a number of internet resources. The National Weather Aviation service provides observations (METAR) and forecasts (TAF) for areas with weather reporting capabilities.

Winds aloft are forecast winds at higher altitudes than the surface for locations throughout the United States. Refer to the Pilot's Handbook of Aeronautical Knowledge for an understanding of the winds and temperatures aloft tables. Winds aloft, too, are important for flight planning and safety.

A typical situation during morning hours is cold air from the night settling, creating calm winds at the surface with the winds aloft (300 to 3,000 feet) at 30 knots. As the surface begins to warm from the sun, the cold surface air starts to warm and rise, allowing the high winds from above to mix and lower to the surface. The wind sheer area in between the high winds above and calm winds below is usually turbulent and can overwhelm aircraft or pilot capabilities. Therefore, it is a dangerous practice to look only at the wind sock for surface winds when there could be strong winds above. Winds aloft must be evaluated for safe flight. *[Figure 5-5]*

During initial solo flights, the wind should be relatively calm to fly safely. As experience is gained, pilot wind limitations can be increased. It is not until the pilot has had dual training in crosswinds, bumpy conditions, and significant pilot in command (PIC) time soloing in mild conditions that pilot wind conditions should approach the aircraft limitations. A safe pilot understands aircraft and personal limitations.

Moisture in the air has a significant effect on weather. If the relative humidity is high, the chance of clouds forming at lower altitudes is more likely. Clouds forming at lower altitudes create visibility problems that can create Instrument Meteorological Conditions (IMC) in which the visibility is below that required for safe flight. The temperature-dew point spread is the basis for determining at what altitude moisture condenses and clouds form. It is important to be particularly watchful for low visibilities when the air and dew point temperatures are within a spread of three to four degrees.

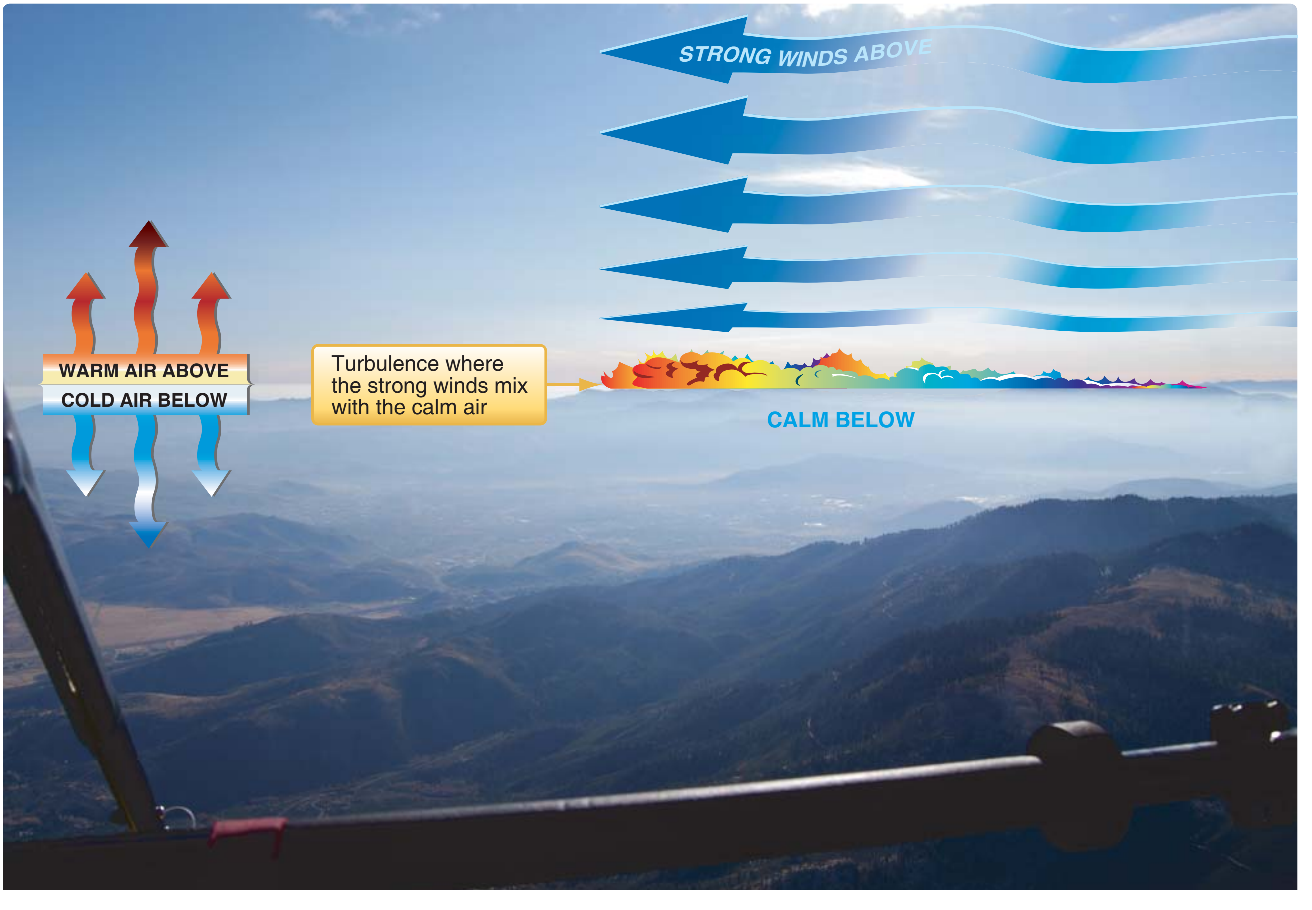


Figure 5-5. *Typical morning inversion layer—calm cold air is below; high winds are above.*

The closer these temperatures are to each other, the greater the chance for fog or clouds forming with reduced visibility conditions. Consider a scenario where the destination airport currently has a temperature-dew point spread of 4 °F, and it is evening when the atmosphere is cooling down. Since the temperature-dew point convergence rate is 4.4° for every thousand feet, the clouds/ceiling would be about 1,000 feet above ground level (AGL). Since it is cooling down, the temperature-dew point spread is decreasing, lowering the cloud level. Therefore, the 1,000 foot AGL ceiling is lowering, creating IMC conditions that are not safe. For this scenario, the flight should not be attempted.

Air temperature and humidity directly affect the performance of the WSC wing and engine. The higher the temperature, humidity, and actual altitude of the operating field, the greater role density altitude plays in determining how much runway the WSC aircraft needs to get off the ground with the load on board, and how much climb performance is required once airborne. The WSC aircraft may have cleared the obstacle at 8 a.m. when the weather conditions were cooler with less humidity; at 1 p.m. with increased air temperature and higher humidity levels, the pilot must reevaluate the performance of that same aircraft. A full understanding of density altitude is necessary to be a safe WSC pilot; refer to the Pilot's Handbook of Aeronautical Knowledge for density altitude and weight effects on performance.

The rate of temperature decrease with increased altitude determines the stability of the air. The stability of the air determines the vertical air currents that develop during the day as the area is heated by the sun. These rising vertical air currents are commonly known as thermals. Generally, stable air has mild thermals and therefore less turbulence than unstable air. Unstable air rises faster, creating greater turbulence. Highly unstable air rises rapidly and, with enough moisture, can build into thunderstorms.

Air stability is easily determined by the rate at which the temperature drops with increased altitude. A standard atmosphere is where the temperature drops 2 °C for every 1,000 foot increase. If the temperature drops less than 2 °C per thousand feet, the air is more stable with less vertical wind (thermals) developed during the day. If the temperature drops more than 2 °C per thousand feet, the air is more unstable with more powerful vertical air currents developed during the day, creating greater turbulence.

In addition to air stability, barometric pressure has a large effect on weather. Low pressure in the area, below the standard atmosphere of 29.92 "Hg, is generally rising air with dynamic and unsettled weather. High pressure above the standard atmosphere in the area is generally sinking air resulting in good weather for flying.

Many airports have automated weather systems in which pilots can call the automated weather sensor platforms that collect weather data at airports and listen to this information via radio and/or land line. Radio frequencies are on the sectional chart and the A/FD has the telephone numbers for these stations. The systems currently available are the Automated Surface Observing System (ASOS), Automated Weather Sensor System (AWSS), and Automated Weather Observation System (AWOS).

Local conditions of wind, moisture, stability, and barometric pressure are factors that should be researched before flight to make a competent decision of go or no go to fly. High winds and moist unstable air with a low barometric pressure indicate undesirable flying conditions. Light winds and dry stable air with high pressure indicate favorable flying conditions.

Pilots should research and document these local conditions before flight to predict the flying conditions and compare the actual flying conditions to the predictions to learn and develop knowledge from the information resources available for flight.

In addition to weather, the National Airspace needs to be checked to ensure there are no temporary flight restrictions (TFR) for the locations planned to fly. TFRs may be found at www.tfr.faa.gov/. For a complete preflight briefing of weather and TFRs, call 1-800-WX-BRIEF.

Clouds visually tell what the air is doing, which provides valuable information for any flight. To understand the different cloud formations and the ground/air effects produced, refer to weather theory in the Pilot's Handbook of Aeronautical Knowledge. *[Figure 5-6]* Cloud clearance and visibility should be maintained for the operations intended to be conducted. The chapter covering the National Airspace System (NAS) provides cloud clearance requirements in each class of airspace. A pilot should not fly when ground and flight visibility are below minimums for his or her pilot certificate and the class of airspace where operating.

Knowledge of mechanical turbulence and how to determine where it can occur is also important. The lee side of objects can feel turbulence from the wind up to ten times the height of the object. The stronger the wind is, the stronger the turbulence is. *[Figures 5-7* and *5-8]*

In addition to adhering to the regulations and manufacturer recommendations for weather conditions, it is important to develop a set of personal minimums such as wind limitations, time of day, and temperature-dew point spread. These minimums will evolve as a pilot gains experience and are also dependent on recency and currency in the make/model of aircraft being flown.

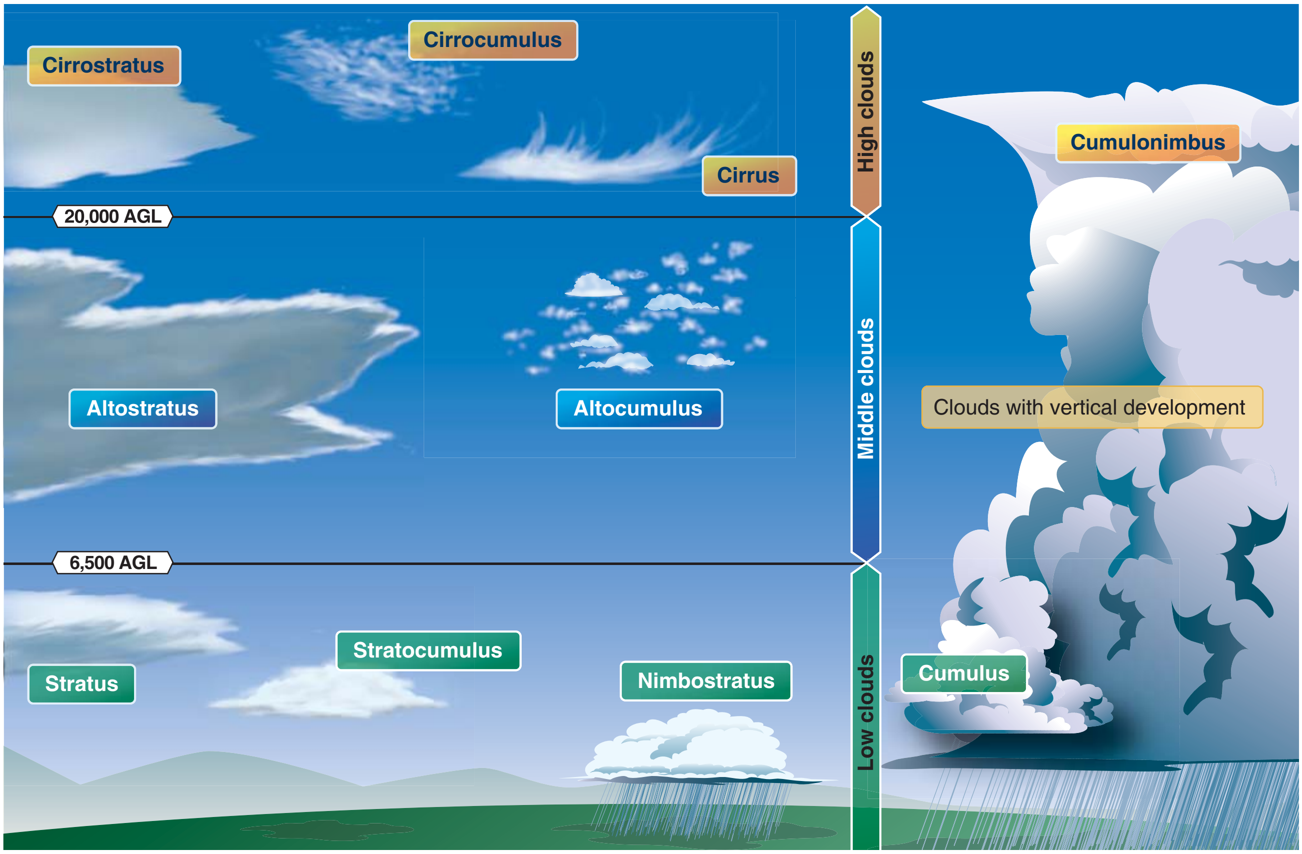


Figure 5-6. *Cloud diagram.*

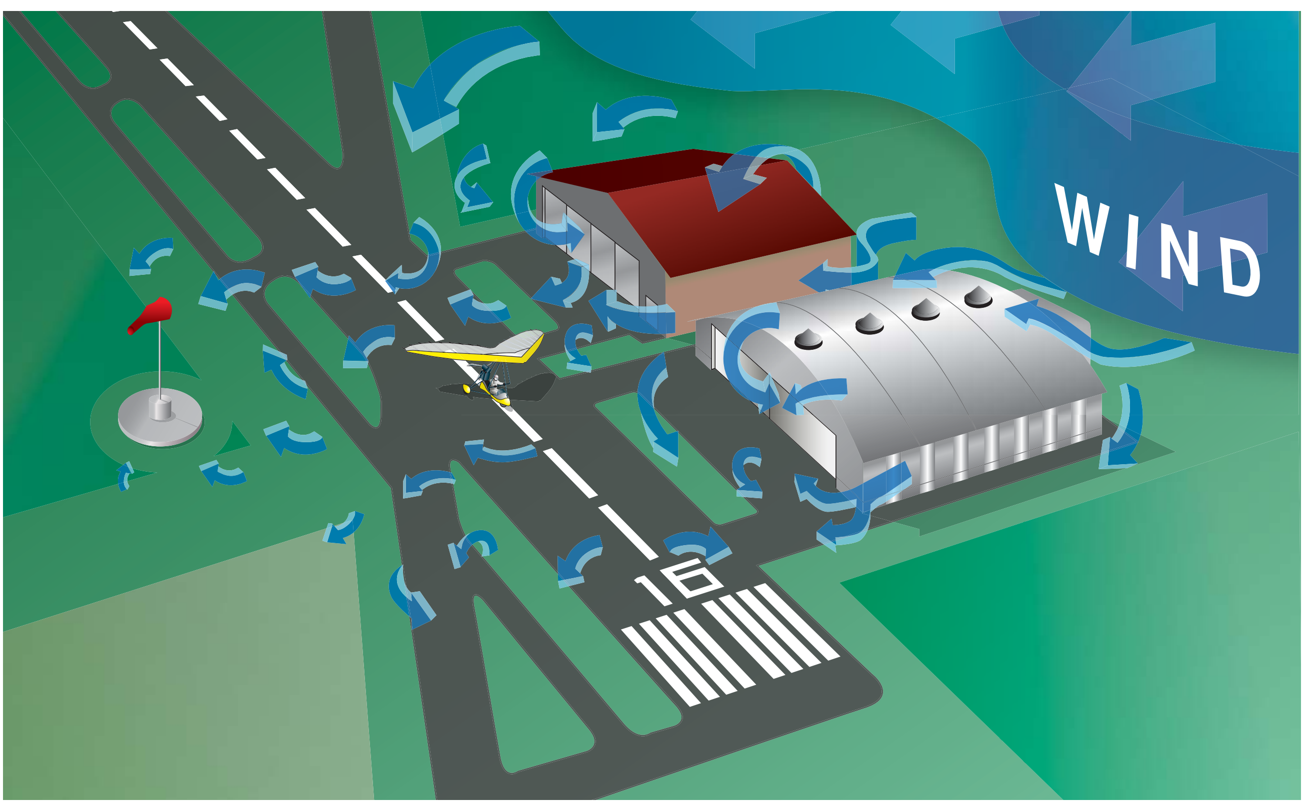


Figure 5-7. *Turbulence created by manmade items.*

Figure 5-8. *Turbulence created by natural land formations.*

Weight and Loading

Weight and loading must be considered before each flight. Do not exceed the maximum gross weight as specified in the pilot's operating handbook (POH). The balance of the pilot, passenger, fuel, and baggage is usually not an issue, but must be reviewed in the POH for the specific make/model since some may have balance limitations. The fore and aft carriage attachment to the wing hang point must be within the limits as specified in the POH for weight and loading of the carriage. Always follow the POH performance limitations.

Transporting

It is best to keep the WSC aircraft in an enclosed hangar, but trailers may be used to transport, store, and retrieve the WSC carriage. If the trailer is large enough, the wing can also fit inside the trailer. If not, then it must fit on top of a trailer, truck, or recreational vehicle (RV). *[Figure 5-9]*

Figure 5-9. *Enclosed trailer containing carriage and wing on top of RV.*

Enclosed trailers are preferred so the carriage is protected from the outside elements such as dust, rain, mud, road debris, and the interested person who may want to tinker with the carriage. The WSC carriage should fit snuggly without being forced, be guarded against chafing, and well secured within any trailer. It is best to utilize hard points on the carriage frame and secure each wheel so the carriage cannot move fore and aft during transport. This is best accomplished by first tying the front wheel from the axles, the fork, or a hard point on the frame with a slight forward pull. Then, secure the rear wheels from the axles or a hard point on the frame with a slight rearward pull. Guides on the side of the wheels and wheel chocks in front and back of each wheel are additionally helpful to secure the carriage on any trailer.

The wing must have ample padding and should have at least three support points where it rests for transport. Transporting the wing properly is of critical importance because the wing resting on any hard surface can wear a hole in the sail and cause structural damage to the tubing. The greatest wear and tear on a wing can occur during transportation. Each support point should have equal pressure—no single point taking most of the load. The wing should be tied down at each attachment point to secure it, but not tight enough to damage the wing. Wide straps are better than thin ropes because the greater width creates less concentrated pressure on the wing at each tie-down point.

Once the loading of the carriage and wing is complete *[Figure 5-9]*, take a short drive, stop, and check for rubbing or chafing of components.

Prior to taking the tow vehicle and trailer on the road, inspect the tires for proper inflation and adequate tread. Ensure all lights are operable, the hitch is free moving and well lubricated, the tow vehicle attachment is rated for the weight of the trailer, and the vehicle and trailer brakes are operable. Avoid towing with too much or too little tongue weight, which causes the trailer to fishtail at certain speeds, possibly rendering it uncontrollable.

Be extremely cautious when unloading the wing and carriage. This is best done with two people since the wing usually weighs more than 100 pounds *[Figure 5-10]* and the carriage

Figure 5-10. *Crane used for one person to lift 110-pound wing on top of RV for transport.*

usually must roll down some incline to get from the trailer to the ground. Some carriages may be tail heavy without the wing, and caution must be exercised, especially moving up and down ramps. Check propeller clearance on the ground when transitioning onto or off of a ramp and propeller clearance going into and out of an enclosed trailer. If the carriage is transported in an open trailer, it should be covered and the propeller secured so it does not rotate/windmill during transport.

Setting Up the WSC Aircraft

Find a suitable area to set up the wing, such as grass, cement, or pavement out of the wind. Inside a large hangar is preferable since wind gusts are not a problem. If setting up outside, align the wing perpendicular to the wind. Most wings set up with the same basic procedure shown in *Figures 5-11* through *5-33,* but the POH should be referenced for the specific WSC aircraft.

Rotate the wing bag so the zipper is facing up. *[Figure 5-11]* Unzip the bag. When setting up the wing, pay close attention to the specific pads, where they are located, and how they are attached for each component of the wing. As shown in *Figure 5-12,* the padding is made specifically for the control frame between the downtubes and the control bar. If every pad is not utilized when taking it down and transporting, there

Figure 5-11. *Wing positioned for setup.*

Figure 5-12. *Wing cover bag unzipped, showing unique padding around control frame corner brackets.*

Figure 5-13. *Assembling control frame.*

will be wear on components with cosmetic and/or structural damage to the wing. The POH may specify where pads go during the setup and takedown. However, when setting up any wing it is a good idea to take pictures, draw sketches, or take notes regarding protective pad location so they can be put back in the proper location during take down.

Assemble the triangular control frame without attaching the wires to the nose. *[Figure 5-13]* Rotate the wing up onto its control frame. *[Figure 5-14]* Place the front wires near the control bar so no one walks on them, remove, and roll up the cover bag. *[Figure 5-15]* Release the wing tie straps that are holding the leading edges together. *[Figure 5-16]* Spread the wing slightly. Remove the pads from the wing keel and kingpost. Note the protective pads still on the wing tips protecting them from the ground during most of the wing set up procedure. *[Figure 5-17]* Continually manage the wing pads and wing tie straps by rolling the pads into the cover bag so they do not blow away. *[Figure 5-18]* If the kingpost is loose, insert it onto the keel to stand upright. If the kingpost is attached, swing it upright. Topless wings have no kingpost. Spread the wings as necessary to keep the kingpost straight up, *[Figure 5-19]* spreading them out carefully and evenly. Do not force anything. Ensure the wires are not wrapped around anything. Separate the right and left battens. Separate the straight battens (for a double surface wing) and set them to the side. Lay out the battens, longest to shortest from the root to the tip next to the pocket they

Figure 5-14. *Rotating the wing onto its control frame.*

Figure 5-15. *Placing the front wires at the control frame.*

Figure 5-16. *Removing the straps holding the two wings together.*

Figure 5-17. *Wings spread slightly to raise the kingpost.*

Figure 5-18. *Pads and wing tie straps neatly rolled into wing cover bag.*

Figure 5-19. *Raising the kingpost and spreading the wings as needed to keep the kingpost upright.*

go into on both sides. Note the protective pads are still on the wing tips so they are protected. *[Figure 5-20]* Insert the battens into the batten pockets, starting at the root and work out to the tip. *[Figure 5-21]* Most batten attachments are double pull. *[Figure 5-22]* Some manufacturers use cord or

Figure 5-20. *Wings spread and battens organized to insert into wings. Note small stepladder holding up keel.*

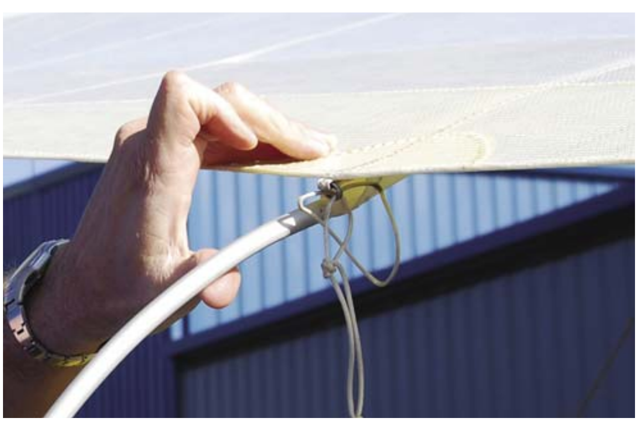


Figure 5-21. *Inserting batten into batten pocket.*

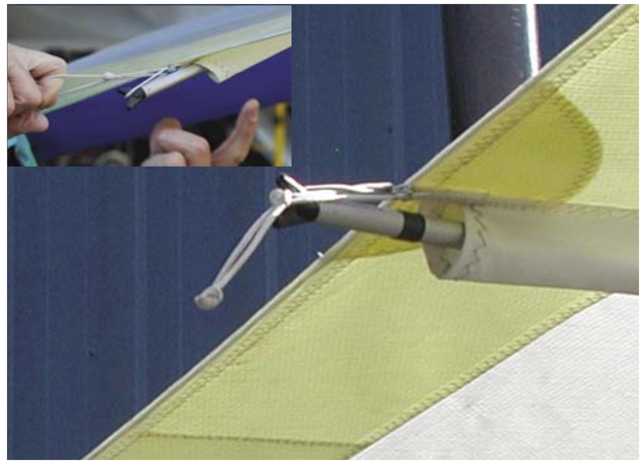


Figure 5-22. *Attaching double pull batten (inset). Batten secured into batten pocket.*

elastic, and others use a system that slips into the sail itself. See the POH for wing details. Insert battens from the root towards the tip about ¾ the way out on each side. Leave the tip battens for later. Spread the wings as far as possible. *[Figure 5-23]* Check to ensure all the wires are straight, not wrapped around, and clear to tension the wing. Tension the wing by pulling back on the crossbar tensioning cable and

Figure 5-23. *Wing ready to tension.*

pulling the crossbar back into position. This may require significant effort for some wings. Secure the tensioning cable to the back of the keel. *[Figure 5-24]* If the keel does not extend out, then support the aft end of the keel to lift the tips off of the ground. *[Figure 5-25]* Move to the front and secure the front control frame flying wires to the underside nose attachment. *[Figure 5-26]* Remove the tip bag protectors and install the tip battens, continuing to move from the root to the tips on each side. Insert the washout strut into the leading edge. Each manufacturer has its own washout strut systems and tip battens. Some manufacturers have no washout struts. Refer to the POH for wing specifics. *[Figure 5-27]*

Figure 5-24. *Attaching the tensioning cables to the back of the wing to complete the wing tensioning step.*

Figure 5-25. *The wing tensioned.*

Figure 5-26. *Attaching the front flying wires to the nose attachment.*

Insert bottom battens for a double surface. If inside a hangar where there is no wind, this can be done by putting the nose down, making it easier to install the lower battens. *[Figure 5-28]* If not already accomplished, lift up on the back of the keel and put the wing on its nose. Lower the undercarriage mast and line up the undercarriage behind the wing exactly in the middle. Move the undercarriage forward and attach the mast to the proper hang point location on the wing keel. Consult the POH for the proper hang point for desired trim, speed, and loading at this time. Attach the backup cable at this time also. *[Figure 5-29]*

Figure 5-27. *Installing the wing tip battens.*

Figure 5-28. *Installing the lower surface battens.*

Figure 5-29. *Attaching the mast to the wing after checking the POH for the proper hang point location.*

Lift up the nose and let the carriage roll backward until the wing is level and the control bar is in front of the front wheel of the carriage. Engage the parking brake and chock the back of the carriage wheels. Ensure everything in the flight deck is free and clear so the wing can be lifted freely into position. *[Figure 5-30]* Lift the wing into position and lock the carriage mast. This position is unique to each manufacturer as some masts hinge above the flight deck. Refer to the POH for details on a specific aircraft. *[Figure 5-31]* Install the carriage front tube. Secure the control bar to the front tube with a bungee. *[Figure 5-32]* Attach any fairing or seats as required. *[Figure 5-33]*

Figure 5-30. *Wing in position and carriage chocked to lift the wing.*

Figure 5-31. *Lifting the wing up into position.*

An alternate method of setting up the wing is to do so on the ground. This is not preferable because the sail is susceptible to getting dirty. However, this method could be used for setting up wings if it is windy or if recommended by a particular manufacturer. The ground method steps are the same as

Figure 5-32. *Attaching the front tube.*

Figure 5-33. *Installing the seats.*

those in the assembly procedure except after the control bar is assembled, the wing is rolled over so the control frame is under the wing. The wing is assembled as if it were standing on its control frame. After the wing is tensioned, the nose is lifted, the control frame pulled forward, and the nose wire secured. This is not a common practice, and the POH should be reviewed for details on this method if it is allowed by the manufacturer.

Taking Down the WSC Aircraft

Find a suitable area to take down the wing, preferably grass, cement, or pavement out of the wind. The best place is in a large hangar so no wind gusts can affect the takedown. If outside, align the wing perpendicular to the wind.

It is important to note that during the take down process, all protective pads must be put in the proper place so that no hardware can rub against the sail or frame during transport. The POH should specify what pads go where. Overall, pad everything along the wing keel plus the kingpost to prevent cosmetic and/or structural damage occurring during transport.

Taking down a WSC aircraft is done in the reverse order of assembly with the following additional steps provided to get the wing neatly packed and organized into the bag. After the wing is detensioned and the battens have been removed from the wing, keep the right and left battens separate for easier sorting during the next assembly.

Carefully bring the wings in towards the keel and pull the sail material out and over the top of the leading edges. Lower the kingpost and pad it top and bottom. This is also the time to pad the area underneath where the control frame is attached to the keel and where the wires are attached to the rear of the keel. *[Figure 5-34]* Bring the leading edges to the keel and keep the sail pulled out over the top of the leading edge, roll it up, and tuck the sail into the leading edge stiffener. Fasten around the leading edge with sail ties. *[Figure 5-35]* It is best to take one sail tie and secure the two leading edges together so it fits into the bag. *[Figure 5-36]* Continue with the reverse order (bag on, flip wing over, and disassemble control frame at downtube and control bar junction). After the control frame is disassembled and laid flat along the wing as shown, the wires are not organized. *[Figure 5-37]* Pull the cables forward towards the nose and organize them so they are straight. Install the protective control frame pads and carefully zip up the bag while tucking everything in so there is no stress on the zipper. *[Figure 5-38]*

Figure 5-34. *Padding the keel and kingpost with the right hand sail over the top of the leading edge.*

Figure 5-35. *Left hand side rolled up and secured with wing tie. Rolling right hand sail which will also be secured with wing tie.*

Figure 5-36. *Securing both leading edges together so the wing easily fits into the bag.*

Figure 5-37. *Control bar folded down along leading edges but wires not yet organized.*

Figure 5-38. *Carefully zipping bag with minimum stress by tucking in wires and organizing components.*

Wing Tuning

Wings are designed to fly straight with a range of trim speeds determined by the manufacturer. If the wing does not fly straight or trim to the manufacturer's specifications, it must be tuned to fly properly. Any wing adjustment can change the handling and stability characteristics of the wing. Each wing is unique and the tuning procedures are unique for each wing. It is very important to follow the specific tuning procedures in the POH/AFM for the specific wing. The following are general guidelines to understand the tuning process.

Tuning the Wing To Fly Straight

Wings may turn to the right or left (depending on which way the propeller turns) at high power settings because of the turning effect described earlier in the aerodynamics section. If it does not fly straight for cruising flight, visually examine for any asymmetric right and left features on the wing before making any adjustments. Look for symmetry in the twist angle. Inspect the leading edge for any discontinuities, bumps, or an irregular leading edge stiffener. Ensure the pockets are zippered and symmetrical on both sides. Ensure the reflex lines are clear, straight, and routed properly. Check the battens to ensure the right and left match (do not make any adjustments in the battens initially because reflex may have been added at the factory initially for tuning), and ensure the battens match the manufacturer's batten pattern. Check the batten tension on both sides and the leading edge tension to ensure it is symmetrical. If it is a used wing just acquired, research the history of the wing to see what might have happened which would cause it to not fly straight. For new wings, contact the manufacturer for advice.

If these checks do not make the wing fly straight, then adjust the twist in the wing according to the manufacturer's instructions. More twist on one side decreases angle of attack, produces less lift, and will drop the wing, which makes it turn in the direction where more twist was added. For example, with an unwanted left hand turn, either decrease the twist on the left hand wing (increase angle of attack at the tip) or increase the twist on the right hand wing (decrease the angle of attack at the tip).

Batten tension is one way of fixing very mild turns. Increasing the batten tension at the tips especially decreases twist and raises the wing. For normal mild turns, most wings have an adjustment at the tip where you can rotate the wing tip around the leading edge. This is the easiest and most effective wing twist adjustment. *[Figure 5-39]* For some models, reflex at the root can be adjusted on a side to adjust a significant turn. More reflex on a side means wing up, similar to reducing twist in a wing. As emphasized above, the POH for each manufacturer must be used for adjusting twist for wing tuning.

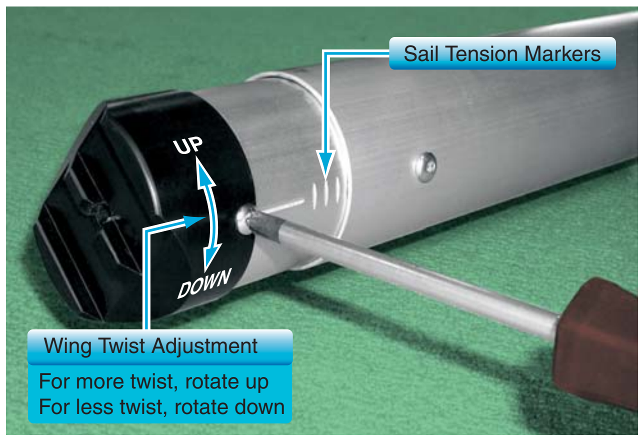


Figure 5-39. *Left hand wing tip twist adjustment shown without sail.*

Adjusting the tension on the leading edge is another method of adjusting the wing twist. However, different wings will react differently when tension is adjusted, so the POH must be followed for a particular wing. Some manufacturers do not suggest adjusting sail tension to adjust twist, but require equal tension with other adjustments to remedy an unwanted turn. For those wings utilizing asymmetrical sail tension to adjust twist, the following information is provided. Adjusting sail tension is most effective on slower wings with lots of twist. Adjusting sail tension affects some high performance wings differently, making it necessary to consult the POH. However, on most wings, increasing sail tension at the tip increases leading edge flex, resulting in more twist.

Tuning the Wing To Fly Slower or Faster

Most wings allow the hang point attachment to move forward to increase trim speed and back to decrease trim speed. If there is a situation where the hang point is at the most forward position and the wing trims below the manufacturer recommended speed, or the trim speed is within 10 miles per hour (mph) of the stall speed, an alternate method for increasing the trim speed is needed. For this situation, the twist must be reduced symmetrically to increase the angle of attack on the tips so they provide more lift and lower the nose for proper trim.

This can be done by pulling back more on the crosstube tensioning cables which reduces the twist in the wing. However, this procedure reduces the stability of the wing and decreases the handling ability of the wing because it is stiffer. This is a common adjustment for hang gliding wings for inflight trim, however this adjustment should only be made on WSC wings as specified in the POH for a specific wing.

Raising and lowering the reflex lines affects airfoil reflex and also changes the trim speed of the wing. Lower reflex lines speed the wing up and make it less stable, raising the reflex lines slows the wing and make it more stable. Some manufactures have this as an adjustable setting which can be varied during flight, other manufactures have this adjustment where it can be made on the ground. Other manufactures do not recommend this adjustment because it can lower the certified stability of the wing.

Preflight Inspection

Each aircraft must have a routine preflight inspection before flight. Use a written checklist during preflight and ground operations to maintain an established procedure. *[Figure 5-40]* A written checklist is required so nothing is forgotten. Ground checklists include preflight preparation, preflight inspection, occupant preflight brief, flight deck management, startup, taxi, before takeoff, and aircraft shutdown. Be smart and follow the regulations—use a written checklist. All checklists should be secured so they do not fly out of the flight deck in flight and hit the propeller. Securing with zippered pockets and having lanyards for the checklists is recommended. Manufacturers of Special Light-Sport Aircraft (S-LSA) have checklists that come with the aircraft. Pilots with an experimental aircraft may need to develop their own.

Figure 5-40. *Laminated index cards are handy for checklists, and sized to fit into the flight suit zippered pocket.*

Certificates and Documents

The first step of preflight inspection is to ensure the aircraft is legally airworthy which is determined in part, by the following certificates and documents:

- **A**irworthiness certificate
- **R**egistration certificate
- **O**perating limitations, which may be in the form of an FAA-approved AFM/POH, placards, instrument markings, or any combination thereof
- **W**eight and balance

ARROW is the acronym commonly used to remember these items. The PIC is responsible for making sure the proper documentation is on board the aircraft when operated. *[Figure 5-41]*

Figure 5-41. *Registration and airworthiness certificates are required to be in plain view.*

Aircraft logbooks are not required to be on board when it is operated. However, inspect the aircraft logbooks prior to flight to confirm the WSC aircraft has had all required inspections. The owner/operator must keep maintenance records for the airframe and powerplant. At a minimum, there must be an annual condition inspection within the preceding 12 calendar months. In addition, the WSC aircraft may also need a 100-hour inspection in accordance with 14 CFR part 91 if it is used for hire (e.g., for training operations). *[Figure 5-42]* If a transponder system is used, the transponder must be inspected within each preceding 24 calendar months.

WSC LSA Maintenance Requirements

S-LSA-certified by FAA accepted ASTM consensus standards

- Annual and 100-hour condition inspection may be performed by:
  - LSA Repairman with Maintenance rating (120-hour course)
  - A&P or FAA certificated repair station
- Maintenance,* repair, and alterations may be performed by:
  - LSA Repairman with Maintenance rating (as authorized by manufacturer)
  - A&P or FAA certificated repair station (as authorized by manufacturer)

E-LSA including:

- Ultralights/trainers transitioned to LSA that meet the criteria of 14 CFR Section 21.191(i)(1)**
- Manufacturer S-LSA kits that meet the criteria of 14 CFR Section 21.191(i)(2) (not amateur built)
- Converted from S-LSA that meet the criteria of 14 CFR Section 21.191(i)(3) (see 14 CFR Section 41.1(b) for servicing)
  - Annual condition inspection may be performed by:
    - > LSA Repairman with Maintenance rating (120-hour course)
    - > A&P or FAA certificated repair station
    - > Owner Repairman with Inspection rating (16-hour course)
  - Owner can be trained in his/her own aircraft and does not need 100-hour inspection.
  - Servicing, repair, and alterations may be performed by anyone.***

Amateur built that meet the definition of LSA and criteria of 14 CFR section 21.191(g)

- Annual condition inspection may be performed by:
  - Original builder gets Repairman certificate for that specific airplane and can perform annual condition inspection:
  - If owner was not original builder, Annual condition inspection may be performed by:
    - > A&P or FAA certificated repair station or original builder
    - > Original builder
- Owner can be trained in his/her own aircraft; 100-hour inspection not necessary
- Servicing, repair, and alterations may be performed by anyone***

* Simple "preventive maintenance" as specified by manufacturer can be done by the owner and operator of a S-LSA with a Sport Pilot or higher certificate.

** 100-hour inspection if used for training, compensation, or hire (if applicable) before January 31, 2010 (towing no end date) may be performed by LSA Repairman with Maintenance rating, A&P or FAA certificated repair station.

*** Maintenance is a common term, but it is not used here because the FAA uses the word "maintenance" to refer to a specific level of service required to be performed by properly trained mechanics.

Figure 5-42. *Maintenance requirements for WSC LSA.*

The pilot must have in his or her possession a Sport pilot certificate for the aircraft being flown, medical eligibility, and a government issued photo identification. For a Sport Pilot Certificate, medical eligibility can be a valid United States driver's license, which also serves as government issued photo identification.

To fly the aircraft with Private Pilot privileges, the pilot needs a valid FAA minimum third class medical certificate accompanied by a government issued photo identification and Private Pilot certificate for WSC aircraft. See Chapter 1, Introduction to Weight-Shift Control, for details on specific pilot certificates and privileges.

Routine Preflight Inspection

The accomplishment of a safe flight begins with a careful and systematic routine preflight inspection to determine if the aircraft is in a condition for safe flight. The preflight inspection should be performed in accordance with a printed checklist provided by the manufacturer for the specific model of the aircraft. However, the following general areas are applicable to all WSC aircraft.

The preflight inspection begins as soon as a pilot approaches the aircraft. Since the WSC aircraft can be transported by trailer, first and foremost, look for any damage that may have occurred during takedown, loading, transit, unloading, and setup. Make note of the general appearance of the aircraft, looking for obvious discrepancies such as tires with low air pressure, structural distortion, wear points, and dripping fuel or oil leaks. All tie-downs, control locks, and chocks should be removed during the unloading process.

The pilot must be thoroughly familiar with the locations and functions of the aircraft systems, switches, and controls. Use the preflight inspection as an orientation when operating a particular model for the first time.

The actual walk-around routine preflight inspection has been used for years from the smallest general aviation airplane to the largest commercial jet. The walk-around is thorough and systematic, and should be done the same way each time an aircraft is flown. In addition to seeing the aircraft up close, it requires taking the appropriate action whenever a discrepancy is discovered. A WSC aircraft walk-around covers four main tasks:

1. Wing inspection
2. Carriage inspection
3. Powerplant inspection
4. Equipment check

Throughout the inspection, check for proper operation of systems, secure nuts/bolts/attachments/hardware, look for any signs of deterioration or deformation of any components/systems, such as dents, signs of excessive wear, bending, tears, or misalignment of any components and/or cracks.

Each WSC aircraft should have a specific routine preflight inspection checklist, but the following can be used as an example and guideline.

Wing Inspection

Start with the nose. Inspect the nose plates and the attachment to the leading edges and keel. Ensure the nose plates are not cracked and the bolts are fastened securely. Check the wire attachments, top and bottom.

Inspect the control frame, down tubes and control bar for dents and ensure they are straight. Inspect the control frame attachment to the keel. Inspect the control bar to down tube brackets and bolts. *[Figure 5-43]* Inspect fore and aft flying wire condition, attachment to the keel, and the lower control bar corner brackets.

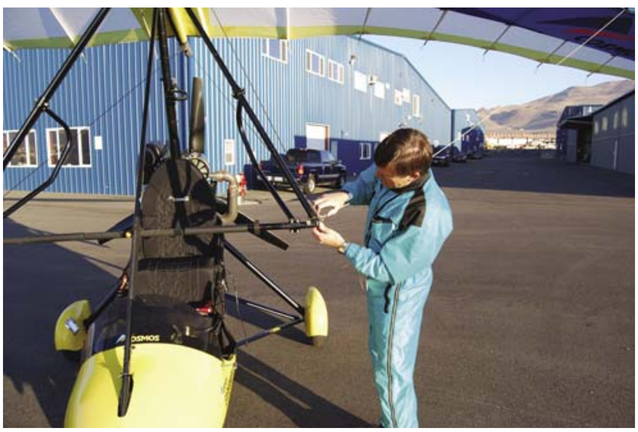


Figure 5-43. *Inspecting the control frame brackets and flying wire attachments.*

Inspect the left side flying wire attachment to the control bar bracket and condition of the flying wire up to the wing attachment. Examine the flying wire attachment to the leading edge and crossbar, as well as all hardware at this crossbar and leading edge junction. *[Figure 5-44]* Inspect the condition of the crossbar and the leading edge from the nose to the tip. Any discrepancies or tears in the leading edge fabric must lead to more detailed investigation of the leading edge spar itself.

Inspect the tip area, including the washout strut and general condition of the tip. If it is a double surface wing, look inside the tip and examine the inside of the wing and its components. *[Figure 5-45]*

From the tip, inspect the surface condition of the fabric. Generally, if the fabric has not been exposed to sunlight for long periods and stored properly, the wing fabric should stay in good shape.

Figure 5-44. *Inspecting the flying wire attachment to the leading edge and crossbar along with all the hardware at this junction.*

Figure 5-45. *Examining inside the tip of the wing to inspect all the components.*

Move along the trailing edge of the wing, inspecting the condition of the trailing edge and the tip batten attachments back to the keel. *[Figure 5-46]* Inspect the sail material, top and bottom, on the wing. Note that the trailing edge is

Figure 5-46. *Inspecting the trailing edge of the wing.*